

When Agent Failure Prediction Measures Task Difficulty Rather Than the Run

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Agent failure predictors report AUROC between 0.85 and 0.94, and this is taken as evidence that a running agent can be monitored and stopped early when it goes wrong. We show the metric answers a different question. Pooled AUROC decomposes exactly into a same-unit and a cross-unit component, and the same-unit component—the one a monitor needs—carries weight $\Theta(1/n)$ —between 0.003% and 0.07% across our corpora. As units grow, pooled AUROC becomes increasingly dominated by between-unit comparisons—which in our data are largely explained by task-difficulty information rather than anything about the run.

Using four corpora that record repeated attempts at the same task by the same model (153,944 trajectories from 21 models in three, plus a 9,989-run probe corpus from two more), we measure the run-level quantity directly. A leave-one-run-out task-difficulty oracle reaches pooled AUROC 0.894; independently annotated task properties alone (difficulty and domain, no model identity) reach 0.826, and adding the entire trajectory to the oracle buys 0.029. Run-level discrimination is weak early—0.51 at one turn, reaching 0.69 only by the end of an episode—and a reimplement of a published predictor reproduces its reported SWE-agent AUROC (0.806 against 0.799) while showing that its fractional-checkpoint protocol leaks trajectory length worth 0.064 AUROC.

These results suggest that runtime failure monitors should be evaluated within task, on prefixes a deployed monitor could compute, and compared against task-level compute-allocation baselines before being used for intervention. A secondary finding—that run-level legibility declines with acting-model capability (Spearman -0.66 , an association)—further narrows when such monitoring can help.

CCS Concepts: • **Software and its engineering** → **Software verification and validation**; *Software testing and debugging*; • **Computing methodologies** → *Machine learning*.

Additional Key Words and Phrases: LLM agents, failure prediction, evaluation validity, measurement, empirical software engineering

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1 INTRODUCTION

LLM agents now do multi-step software-engineering work—and fail often, at measured rates of 41–86.7% across seven frameworks [3], each failure costing minutes and many tokens. Hence a growing literature reads the partial execution trace of a running agent and scores impending failure, reporting strong numbers: verbal-confidence AUROC 0.85 at completion [17], hidden-state probes said to anticipate failure from the first round while saving 54–60% of tokens [24], automaton monitors up to 0.941 [5]—a literature spanning ICML, NeurIPS, ICLR, EMNLP and arXiv.

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Three observations that do not fit. Three published findings sit awkwardly beside those numbers.

First, accuracy does not translate into deployment benefit: a critic with offline AUROC 0.94, used to intervene in running agents, made end-to-end performance *worse*—up to 7.3 disrupted successes per recovered failure [31]. **Second**, a model that never observes the agent does about as well: predicting success *before the run starts*, from static task properties alone, reaches AUROC 0.841 on held-out SWE-bench Verified [11]—inside the range reported by systems reading entire trajectories, in a literature that does not cite them. **Third**, much of the outcome is not a property of the task at all: the same agent on the same task yields one success and one failure roughly a fifth of the time [16], and at the moment such a pair begins the two runs are identical in every observable respect.

What we think is going on. Consider a forecaster evaluated across Seattle and Phoenix. “Rain in Seattle, sun in Phoenix” scores well without knowing anything about weather—and asked whether it will rain in Seattle *tomorrow*, it offers nothing.

The metric the failure-prediction literature reports has this structure. It pools runs from many different tasks and asks whether failures rank above successes overall. Failures come disproportionately from hard tasks. A predictor that estimates only task difficulty therefore scores well without having learned anything about any individual run—and run-level discrimination is precisely what aborting a run requires. Hence:

A predictor that cannot separate two attempts at the same task knows only the task. It can justify moving effort between tasks; it cannot justify aborting the attempt in front of it on that attempt’s evidence.

Software engineering has met both halves of this problem. The pooled/within distinction is the agent-monitoring form of cross-project versus within-project defect prediction [40], and the lesson that a predictive metric must match the decision it informs is itself an SE-native one [23], as is the concern that accuracy misleads on the imbalanced data typical of defect prediction [8]; mixed-outcome units are the agent-run analogue of flaky tests [19]—our 22–41% mixed rates are flakiness rates for agent runs, and a flaky unit’s outcome cannot be read off the task’s identity any more than a flaky test’s verdict can be read off the test’s.

1.1 What is already known, and what is not

Three layers stay distinct. *Known*: pooled ROC can differ from covariate-specific discrimination; biostatistics developed covariate-adjusted ROC curves for this [13, 22]. *Ours, diagnosis*: in repeated agent benchmarks the same-task pair weight collapses as $\Theta(1/n)$, so failure-prediction evaluations are dominated by cross-task comparisons, published monitors included. *Ours, consequence*: strong pooled AUROC can mean weak support for the decision the monitor exists to make, and selecting policies by it selects worse ones.

Two recent studies already report that early failure prediction does not work (Li et al. [17] via path switching at 50% progress; Lee [16] via canonical-path deviation on mixed-outcome units, whose identification strategy we adopt directly); §3.2 details both. We claim no novelty there. Our claims are scoped to prefixes—the intervention regime—and the open question is metrological: what do the high reported accuracies measure?

Neither study decomposes the reported accuracy. That decomposition is what this paper supplies, together with its consequences: a diagnostic, a reproduction on the strongest available features, and a replacement policy that outperforms the recommendation Li et al. [17] derive from their own findings.

Some benchmarks record several independent attempts at the same task by the same model; within such a unit, task and model are fixed by construction and only the run varies. Measuring

accuracy *inside* units isolates run-level discrimination; comparing it with the pooled figure separates the questions the pooled metric conflates.

1.2 Research questions

RQ1 *What does pooled failure-prediction accuracy measure in benchmarks that record repeated attempts?*

RQ2 *How much run-specific information is available at online-computable prefixes—across predictors, models, corpora, and hidden-state probes?*

RQ3 *Does evaluating the correct estimand change intervention and resource-allocation conclusions?*

The allocation policy of §6 presupposes prior attempts at the task by some model; it targets repeated-evaluation settings—benchmark runs, CI re-execution, model migration—rather than a first attempt at a novel issue.

Within RQ2 the hidden-state probe is the strongest objection to a behavioural null: a reader may grant that visible behaviour carries no early run-level signal and still hold that it is latent in the acting model’s activations—the premise of an active probing literature—so we test it on the model that produced the trajectories.

1.3 Contributions

- **Estimand analysis.** We give the exact pairwise decomposition $A_{\text{pool}} = w A_{\text{within}} + (1 - w) A_{\text{cross}}$ and show the same-task weight $w = \Theta(1/n)$ is vanishing in realistic benchmarks—between 3.4×10^{-5} (C2) and 6.8×10^{-4} (C4-L), so at least 99.93% of the pairs behind the pooled figure compare different units. The pooled-vs-conditional distinction is established statistics [13, 22]; its extreme, load-bearing form in agent evaluation is the contribution (§2.3).
- **Empirical diagnosis.** Across four corpora, 21 models, three scaffolds (153,944 trajectories in C1–C3, plus a 9,989-run probe corpus from two further models): a leave-one-run-out difficulty oracle reaches pooled 0.894 (0.909 on non-coding tasks alone) and task metadata alone 0.826 (model identity alone only 0.568), the full trajectory adds +0.029, and early within-task discrimination sits at or near our resolution boundary (§5.1–§5.3).
- **Validation on published protocols.** A reimplemented published monitor reproduces its reported SWE-agent pooled AUROC (0.806 vs 0.799) yet is far weaker within task early (0.507 at one turn; 0.688 at full trajectory), and its fractional checkpoints leak eventual length worth +0.064 at matched observation depth (§5.9).
- **Deployment consequences.** At equal budget in replay (Table 6), aborting on a monitor of the measured quality costs 19% of solved tasks at the largest budget, adding it to an allocation policy hurts at every studied budget, and allocation solves up to 3.6× more issues than round-robin; the supplement’s abort-and-restart simulation shows the pooled-favoured policy harming end-to-end success (§6.7).

Secondary findings delimit these claims: signal accumulates with depth on a fixed cohort; legibility declines with acting-model capability ($\rho = -0.66$; within families -0.75); and behavioural uniformity, though it rises with capability, does not mediate that decline (§5.10).

2 HOW TO READ THE MEASUREMENTS IN THIS PAPER

This section defines every quantity we use before we use it. Readers familiar with AUROC may still want §2.2, which introduces the paper’s central distinction.

Runs, units, and mixed-outcome units. A *run* is one complete attempt at a task, labelled success or failure by the benchmark’s oracle. A *unit* is a (model, task) pair; repeated runs of one unit hold task

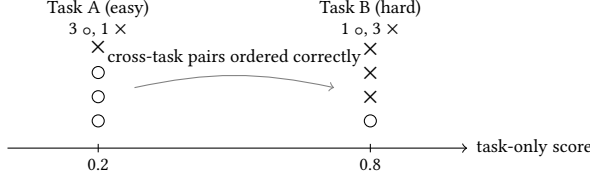


Fig. 1. A score knowing only task identity: cross-task pairs rank correctly (pooled AUROC 0.75 here) while every within-task pair ties (within-task exactly 0.5); with n tasks, cross-task pairs dominate by $\Theta(n)$ (§2.3).

and model fixed, so only the run varies. A unit is *mixed* when it has at least one success and one failure; only mixed units carry information about A_{within} , a locality we analyse in §8.

AUROC has the pairwise reading we use throughout: pick one failed and one successful run at random; AUROC is the probability the predictor scores the failed run higher (ties count half). 0.5 is a coin flip.

2.1 Three questions, three estimands

One number is routinely treated as the answer to three different questions.

Q1: cross-task triage (*which tasks will fail?*): task-level prediction; pre-execution difficulty estimation (§6.6) serves it. **Q2: run-level monitoring** (*given this task, which attempt will fail?*): the question a monitor answers when it aborts the run in front of it; within-task discrimination is its estimand, and it is this paper’s subject. **Q3: benchmark-wide ranking** (*can a score rank all successes above all failures?*): pooled AUROC answers this faithfully—the literature routinely answers Q3 while motivating the work by Q2.

In practice: triage $\rightarrow$ task-level prediction; ranking $\rightarrow$ pooled AUROC; *this-attempt* decisions $\rightarrow$ within-task AUROC on deployable prefixes; spending the next attempt $\rightarrow$ solved-under-budget.

2.2 Pooled versus within-task AUROC

The distinction that organises this paper is *which pairs* of runs the probability above ranges over.

Pooled AUROC, written A_{pool} , draws the failed and successful run from anywhere in the dataset. The two runs will usually come from different tasks. This is the number reported throughout the literature.

Within-task AUROC, written A_{within} , draws both runs from the *same unit*. We compute it per unit and average, weighting each unit by how many success/failure pairs it contributes:

$$A_{\text{within}} = \frac{\sum_u n_u A_u}{\sum_u n_u}, \quad n_u = |S_u| \cdot |F_u|, \quad (1)$$

where S_u and F_u are the successful and failed runs in unit u , and A_u is the AUROC computed within that unit.

Why this matters: a unit fixes task and model, so a predictor that has learned only “this task is hard” scores the same on every run in the unit—exactly 0.5 on A_{within} ; anything above 0.5 must reflect the individual run. We call $A_{\text{pool}} - A_{\text{within}}$ the **gap**, a diagnostic of between-task inflation—deliberately not an *attribution*, since the two are different pairwise estimands; what decomposes exactly is A_{pool} itself, next.

2.3 An exact decomposition of pooled AUROC

Pooled AUROC is a probability over positive–negative pairs, which partition into same-unit and cross-unit pairs. Writing w for the share of pairs that are same-unit, this partition gives an identity:

$$A_{\text{pool}} = w A_{\text{within}} + (1 - w) A_{\text{cross}}, \quad w = \frac{\sum_u P_u N_u}{(\sum_u P_u)(\sum_u N_u)},$$

where P_u and N_u count the failing and succeeding runs of unit u , and A_{cross} is concordance over cross-unit pairs. This holds by construction for any scoring function; we verify it numerically to 10^{-9} in every configuration we report.

The identity matters because of how small w is. For n units of r runs each, the same-unit pairs number $O(nr^2)$ against $O(n^2r^2)$ cross-unit pairs, so $w = \Theta(1/n)$. On C4-L, with 371 units of which 142 are mixed, $w = 0.0007$: 2,259 within-unit pairs against 3,314,835 cross-unit, so $A_{\text{pool}} = 0.0007 \times 0.5241 + 0.9993 \times 0.6313 = 0.6313$ at $k=1$, layer 20—99.93% of pooled AUROC is determined by comparisons *across* tasks; the component answering the deployment question carries 0.07% of the weight and cannot move the reported number regardless of its value.

The identity also yields a *predictive* theory of the confound. With μ the mean failure rate and MD the Gini mean difference of the unit rates q_u , the difficulty oracle’s pooled AUROC has closed form $\frac{1}{2} + \text{MD}/(4\mu(1 - \mu))$ and the same-unit weight the constant $w = \overline{q(1 - q)}/(n\mu(1 - \mu))$ —both computable from a benchmark’s *outcome table alone*. The ceiling predicts 0.954/0.987/0.984/0.973/0.979 for our five corpora/splits against freshly measured leave-one-run-out oracles of 0.945/0.974/0.903/0.949/0.959; and the attenuation is itself derivable—a drawn run’s leave-one-out score is $\text{Bin}(r-1, q_u)/(r-1)$ exactly, giving a finite-run formula that roughly halves the ceiling’s error on every corpus (0.950/0.980/0.940/0.962/0.969; residual falls with runs per unit, from 0.037 at C3’s three to 0.005–0.013 at ten or more, its sign and size consistent with plug-in dispersion bias). Derivations in the supplement; the w predictions match §4.1 exactly. Larger benchmarks make the mismatch worse: w falls as $1/n$ while the ceiling depends only on difficulty spread. Formally, A_{within} is a member of the covariate-adjusted ROC family [13] with covariate = unit; it is not identical to their estimator—they weight covariate-specific AUCs by the case distribution, we by pair counts $P_u N_u$ —and the agent-specific content here is the $\Theta(1/n)$ collapse, the ceiling formula, and their deployment consequences, not the family.

2.4 Variance decomposition and deployment accounting

The intraclass correlation (ICC) [21] is the share of outcome variance lying *between* units, $\text{Var}(\bar{y}_u)/\text{Var}(y)$; its complement bounds what any predictor reading only the current run can explain. For the deployment analysis (RQ3) we adopt the accounting of Vasudev et al. [31]: with recovery rate r (restart succeeds given the original would fail) and disruption rate d (restart fails given the original would succeed), intervening pays only when the baseline failure rate exceeds the break-even $d/(r + d)$.

2.5 Uncertainty and hypothesis tests

Confidence intervals. All intervals are 95% percentile bootstrap intervals computed by resampling *units*, not individual runs, with 2000–5000 resamples. Resampling runs would understate uncertainty, because runs inside a unit are correlated by construction.

Early-prefix hypothesis tests. For analyses asking whether useful run-specific discrimination is already present at intervention-relevant prefixes, we test $H_0: A_{\text{within}} = 0.5$. This is not the paper’s central claim—the pooled/within decomposition holds for any predictor and any value of A_{within} , and our own data show substantial run-level signal late in episodes—but it quantifies how much

signal is detectable where an intervention could still save compute. A non-rejection is not proof of absence: for the C4 probe analyses specifically, the power calculation of §5.7 and the label-noise calibration of §5.8 place the effective resolution near $A_{\text{within}} \approx 0.55$ (effects around 0.53 cannot be excluded; effects near 0.60 would be readily detected), so we describe early probe discrimination there as *weak* rather than absent. Testing demands more than a confidence interval, so we use exact permutation. Inside each unit we shuffle the outcome labels among that unit's runs, holding the number of successes and failures fixed, and recompute A_{within} . Under the null the labels are exchangeable within a unit, so this yields the exact conditional null distribution of the statistic. We use 2000–3000 permutations and report a two-sided p -value.

This is a stronger instrument than a bootstrap interval: it conditions on the observed unit structure, so it accounts for units of unequal size and unequal success counts without any distributional assumption.

Power, comparisons, corrections. Beside each p -value we report the permutation null's standard deviation (twice it is roughly the smallest detectable deviation from 0.5); this SD also has a closed form, $\sqrt{\sum_u P_u N_u (P_u + N_u + 1) / 12} / \sum_u P_u N_u$ from the exact Mann–Whitney null variance, which reproduces the empirical values (0.0058 vs 0.0057 on C2; 0.0183 vs 0.0177–0.0190 on C3; 0.020 vs 0.021 on the C4-L probe subset)—so power for any repeated-run benchmark is predictable from its outcome table too (supplement). Predictor comparisons bootstrap the *difference* on shared resamples; effect sizes are Cliff's delta; families of within-task tests are corrected with Holm–Bonferroni at $\alpha = 0.05$.

3 BACKGROUND AND RELATED WORK

Two literatures that do not meet. Two camps report the same metric while answering different questions.

Task-level prediction estimates the probability of success from static properties of the task before any execution, for benchmark design, curriculum construction and triage [11, 14]. It is openly a model of task difficulty: Ge et al. [11] explicitly take the “majority ($\geq 50\%$) outcome” per agent–task pair, deliberately collapsing run-to-run variation.

Trajectory-level prediction estimates the same probability from the unfolding run, for monitoring and intervention [5, 17, 24, 36]. A related strand characterises the behavioural correlates of success without building a predictor from them [2, 20, 39], establishing trajectory-level analysis as an SE problem in its own right.

By direct reading, Ge et al. [11] cites no trajectory-based failure prediction and the trajectory papers we surveyed evaluate only against each other: we found no work reporting the control condition “ignore the run, estimate task difficulty.”

A larger body of work localises failures *after* a run completes [6, 32, 37, 38], with low accuracy (best: 53.5% agent-level, 14.2% step-level on Who&When); it is complementary, since post-hoc attribution cannot intervene in a run that has ended.

3.1 Probing the agent's internal state

Closest to our hidden-state analysis is Silva et al. [28], whose corpus we use as C4: probing coding-agent hidden states, they show internal representations encode current and future program properties (up to 0.83 AUC, anticipating edits many steps ahead). Our question is different and complementary: not whether hidden states encode program state, but whether they separate eventual success from failure *among repeated attempts at the same task* at deployment-relevant prefixes. Rich program-state information and near-chance within-task outcome discrimination

turn out to coexist—our positive controls decode length and repository identity from the very activations whose outcome signal is at chance.

The strongest published claim against a behavioural null is Ruan et al. [24]: per-round hidden-state probes said to anticipate failure from the first interaction round, with behavioural scorers “barely better than chance”, and an abort cascade built on top. The claim is evaluated pooled. §5.5 shows on two acting models that the same probe class attains within-task 0.509 and no advantage over a three-feature length baseline; both observations are consistent under our decomposition, because activations encode task difficulty (length is recoverable at R^2 up to 0.92 on C4-Q and 0.98 on C4-L)—sufficient for a pooled score, insufficient for the abort decision. The test requires the acting model’s public weights, which is what C4 provides.

Cost-aware early stopping. A parallel line treats aborting as a decision problem, halting a run when continuation is judged unprofitable [7, 24, 27]. The savings these systems report are real; our contribution is a control condition. A policy conditioning only on the task’s historical failure rate—reading no trajectory—dominates the trajectory-based policies we test at every studied budget in replay (§6), and the classical index-policy result for costly search [34] does not transfer, since our arms retire on success and probabilities are estimated.

3.2 Negative results on early prediction

Two recent studies report that early prediction does not work. Li et al. [17] find that no uncertainty signal exceeds AUROC 0.60 at 50% trajectory progress on deep-research tasks, attributing this to *path switching*. Lee [16] exploit repeated runs on a tool-use benchmark and find the deviation signal statistically indistinguishable from zero at 10%, 25% and 50% of trajectory. Notably, these two studies appeared six months apart, in different domains, without citing each other.

Our work is closest to Lee [16], who introduced the use of mixed-outcome units for causal identification. We adopt their identification strategy unchanged and claim no novelty for it. The questions differ: Lee asks what *causes* agents to fail and answers with a mechanism (deviation from a canonical solution path); Li et al. [17] asks the same and answers with another (path switching). We ask what failure *predictors* measure. A mechanism explains why the signal is absent early; it does not explain why reported accuracies are nonetheless high, which is a property of the evaluation rather than of the agent.

Two differences matter for what can be concluded. First, Lee reports a *positive* within-unit effect at trajectory completion, so run-level signal is not absent in general; our claims concern prefixes, where intervention is still possible, and we do not dispute the completion result. Second, neither study evaluates a probe of the acting model’s internal state, which is the strongest objection to a behavioural null and which §5.5 addresses.

We also arrive at a different practical recommendation. From their findings Li et al. [17] advise using final-step confidence to decide whether to restart, in preference to in-trajectory intervention. We agree that in-trajectory intervention is unsupported, but §6 shows that a policy conditioning on task difficulty alone—requiring no trajectory signal at any point—outperforms every trajectory-based policy we tested in replay.

3.3 A gap in the literature

Four Google Scholar query families (2023 onward), every returned trajectory-level failure-prediction result read and snowballed; each load-bearing claim verified against the cited paper’s own text. The citation-knee stopping rule [26] is inapplicable to a literature whose 2026 works all have zero citations; protocol and verification discrepancies in the supplement.

Table 1. Literature review results. *Pfx*: operates on a partial trajectory. *Rep*: uses repeated runs of the same model on the same task. *Nat*: failures occur naturally rather than being injected. ≥ 5 : evaluates on five or more environments. *Dep*: measures the effect of acting on the prediction. $\sim$ denotes partial satisfaction; n.r. not reported.

Paper	Venue	Pfx	Rep	Nat	≥ 5	Dep
Who&When [38]	ICML'25	x	x	✓	x	x
MAST [3]	NeurIPS'25	x	x	✓	✓	x
TRAIL [6]	arXiv'25	x	x	✓	x	x
AgentRewardBench [18]	arXiv'25	x	x	✓	✓	x
AgenTracer [37]	ICLR'26	x	x	x	✓	$\sim$
DRIFT [32]	arXiv'26	x	x	✓	x	x
Argus [15]	arXiv'26	x	x	✓	✓	x
Psychometrics [11]	arXiv'26	x	$\sim$	✓	✓	x
Failure as Process [39]	arXiv'26	$\sim$	x	✓	x	x
Automata [5]	arXiv'26	✓	x	✓	✓	x
Doomed [24]	arXiv'26	✓	✓	✓	x	$\sim$
Last Step [17]	EMNLP'26	✓	$\sim$	✓	x	$\sim$
Capable [16]	arXiv'26	✓	✓	✓	x	x
Paradox [31]	arXiv'26	✓	n.r.	✓	x	✓
This paper		✓	✓	✓	✓	✓

The retrieved papers divide on five properties, each checkable from a method section and each load-bearing for whether a predictor could be deployed as a monitor (Table 1).

Individual prior works satisfy several of these properties—broad environment coverage and deployment measurement each appear in the table—but none simultaneously combines prefix-level prediction, repeated attempts, natural failures, five-plus environments and deployment evaluation; among the six prefix-level works, three have repeated runs and none has either of the last two properties (Table 1).

One entry deserves comment. Ruan et al. [24] collect 8 rollouts per task (100 TextCraft tasks; 200 WebShop at 4) and split grouped by task—precisely the design the within-task estimand requires. What is reported is pooled AUROC; the within-task quantity is never computed. The pattern this paper is about is visible in the work best placed to avoid it.

This gap is not for want of study—we count more than twenty attribution papers, and two groups independently published the same negative result six months apart without citing each other. Within our search, the area is active in *topic* space yet sparse in the deployment-relevant *property* space of Table 1.

4 STUDY DESIGN

4.1 Corpora

We need benchmarks recording several independent attempts at the same task by the same model. Three public corpora with many runs per unit qualify (Table 2); a fourth, built for the probe requirement, is §4.2. C3's 6,834 runs parse to 6,727 (2,269 units; 500 mixed on all records, 488 among parseable ones—the permutation-test set). The corpora differ in scaffold, benchmark, model family, domain and base failure rate, which is what lets the cross-corpus question be answered. A unit is a (model, task) pair; a unit is *mixed* when it contains at least one success and one failure, and only mixed units inform A_{within} .

Table 2. The three corpora. All are publicly released under permissive licences.

	C1	C2	C3
Source	SWE-agent	SWE-rebench	LiveClawBench
Scaffold	SWE-agent	OpenHands	OpenClaw
Domain	code	code	10 domains
Trajectories	80,036	67,074	6,834
Tasks	3,591	6,306	134
Models	3 (Llama-3 8B–405B)	1 (Qwen3-Coder-480B)	17
Failure rate	0.833	0.521	0.582
Mixed-outcome	27.2%	27.6%	21.9%
Same-unit weight w	2.5×10^{-4}	3.4×10^{-5}	8.8×10^{-5}
Mixed units	926	1,741	500
Within-unit pairs	220,055	38,576	1,000

The mixed-outcome rate—the fraction of (model, task) units that produce both a success and a failure—is between 21.9% and 27.6% in all three corpora despite their differences. Roughly one unit in four is genuinely undetermined by the task and model alone.

C3 additionally carries *independently annotated* difficulty tiers and domain labels—author-produced, not outcome-derived, hence a stronger instrument than any oracle; their validity shows in failure rising monotonically $0.313 \rightarrow 0.710 \rightarrow 0.962$ across tiers.

4.2 A fourth corpus, for the activation probe

The hidden-state probe requires the *acting* model’s weights: hidden states are model-specific, and most public corpora record proprietary-model trajectories.

We use the LATENT PROGRAMMING HORIZONS corpus of Silva et al. [28] (C4), which records SWE-bench trajectories from two openly released mixture-of-experts models: LAGUNA-XS.2 (40 layers, $d_{\text{model}} = 2048$, 256 experts, $\approx 33\text{B}$ total and $\approx 3\text{B}$ active parameters) and QWEN3.6-35B-A3B. Each record stores the *exact token identifiers* the model consumed, together with per-message segment offsets. Teacher-forced replay is therefore exact by construction: we do not reconstruct a chat template and hope it matches, which is a common and undetectable source of error in probe reproductions.

Both splits record ten attempts per task on 500 SWE-bench tasks; mixed-outcome rates (40.6%/34.8%; 203/174 mixed units) exceed C1–C3; C4-L’s median trajectory is 35,759 tokens.

4.3 Probe construction

We cut after assistant turn $k \in \{1, 3, 5, 10\}$, take the hidden state at the last emitted token from layers $\{10, 20, 30, 40\}$ (one causal forward pass to the deepest cut yields all shallower ones), and probe with ℓ_2 -regularised logistic regression on the raw 2048-dim activation—linear deliberately, since linear decodability is the probing literature’s own standard for explicit representation.

Every fit uses GROUPKFOLD by task, so no task appears in both training and test folds and the probe cannot succeed by memorising task identity. The regularisation strength is selected by nested GROUPKFOLD inside each training fold, not on test data (test-set tuning would inflate the apparent activation advantage several-fold).

Prefix lengths at $k=10$ reach 40,133 tokens; we do not truncate below the maximum, since a length cap induces outcome-dependent selection (§4.5).

Table 3. Decomposition on C1 (SWE-agent). Absolute-turn design, confound removed. The oracle row uses only each unit’s failure rate—no run-level information—and covers all 80,036 records; featurised rows cover the 79,929 parseable trajectories.

Condition	n	A_{pool}	A_{within}	gap	units
$k = 2$	79,929	0.6435	0.5169	+0.127	926
$k = 5$	77,951	0.6863	0.5428	+0.144	920
$k = 10$	63,657	0.7219	0.5987	+0.123	885
$k = 20$	35,554	0.7463	0.6527	+0.094	620
Difficulty oracle	80,036	0.9454	0.5000	+0.445	926

4.4 Predictors

Predictors. Trajectory predictors read the first k turns: behavioural features (turn counts, message/observation lengths and dispersion, error tokens, repetition, tool diversity) plus TF-IDF text, fit by logistic regression and gradient boosting, out-of-fold under GROUPKFOLD on task. Label-only predictors (C3) use pre-execution categorical facts—difficulty tier, domain, model—and never observe the run. The difficulty oracle scores each run by its unit’s failure rate, leave-one-out for pooled figures.

One caution: a leave-one-out group mean must not be used for the *within*-task figure—inside a unit its only variation is the subtraction of the run’s own outcome, producing a spurious $A_{\text{within}} = 0$; we use the plain group mean there, which is constant within unit and yields exactly 0.5, as it should.

4.5 Prefix definition, and a confound

A monitor observes a run turn by turn without knowing how long it will last. Our own initial design featurised a fixed *fraction* of each trajectory and included k/T , where T is the final turn count—unknown at monitoring time and outcome-correlated. On C1 that single feature inflates A_{within} by 0.044–0.074 depending on depth, and removing the remaining length features on top changes little, so it acts alone. All results in this paper use absolute turn indices with the feature removed; §5.9 measures the same leak in a published protocol.

5 RESULTS

Where each RQ is answered: RQ1 §5.1 (task-only 0.826, oracle 0.945, both at $A_{\text{within}}=0.5$); RQ2 §5.2–5.3, 5.4, 5.5, 5.9, 5.10 (online signal weak early, real late; probes at chance; controls pass); RQ3 §6.7 (metric choice flips the deployable policy).

5.1 What pooled AUROC measures

On C1 (Table 3), early run-level discrimination is near chance (0.517 at two turns, 0.543 at five), modest only late and only on the 44% of runs surviving twenty turns; the gap is +0.09 to +0.14 throughout.

The oracle row makes the ceiling concrete: knowing only each unit’s failure rate, and nothing about the run, reaches pooled 0.945 at chance within task. And this is not mere oracle leakage: a *deployable* pre-execution predictor—issue-text TF-IDF plus repository identity, tasks held out, no target-task outcomes—reaches pooled 0.598 (C1) and 0.616 (C2) while still scoring 0.500 within task.

Table 4. C3 (LiveClawBench): 17 models, 134 cases, 10 domains. The upper block lists predictors that never observe the run; their within-task AUROC is 0.500 by construction, since they are constant inside a unit. Intervals are 95% bootstrap over units.

Predictor	A_{pool}	A_{within}	gap
<i>never observes the run</i> (held-out, cross-fitted by task)			
model identity only	0.5676	0.5000	+0.068
task only (difficulty $\times$ domain)	0.8259	0.5000	+0.326
task $\times$ model	0.8449	0.5000	+0.345
<i>reads the trajectory</i>			
trajectory, $k = 2$	0.7386	0.4979 [0.464, 0.532]	+0.241
trajectory, $k = 5$	0.7320	0.5415 [0.505, 0.579]	+0.190
trajectory, $k = 10$	0.7521	0.5251 [0.483, 0.567]	+0.227

5.2 Does reading the trajectory help?

C3 answers this most cleanly: its labels are independent annotations, so we lead with them and treat the leave-one-run-out oracle strictly as a ceiling (held-out pre-execution prediction reaches 0.841 on SWE-bench Verified [11], matching our task-only 0.826).

The result is stark, and it is a property of the *task*, not the model. Task metadata alone—difficulty tier and domain—reaches pooled AUROC **0.826** (within-task 0.5); adding model identity lifts it only to 0.845, and model identity alone reaches just 0.568. Removing the model confound entirely, within *each* of the 17 models separately the per-model task oracle reaches pooled median 0.985 (range 0.962–0.998): holding the model fixed, task difficulty alone still exceeds the 0.85–0.94 band. Trajectory predictors top out at 0.732–0.752 pooled, within-task 0.498–0.542 (covering 0.50 at two of three depths). Pooled AUROC would select a predictor with no demonstrated ability to do a monitor’s job.

The variance ceiling. On C3 the ICC is 0.800: only 20% of outcome variance remains within task–model units. This does not mean outcomes are 80% predetermined, and it is not an AUROC bound—but it says most of the variation a pooled evaluation sees is between-unit variation, which is why a label-only predictor does so well on it. The direct measurement of what a predictor gains follows.

5.3 What does reading the trajectory add?

The direct measurement: three nested cross-fitted predictors on C3. M_0 knows only the task (its unit’s failure rate from that unit’s *other* runs); M_1 only the trajectory; M_2 both. Pooled AUROC/Brier/log-loss: M_0 0.894/0.103/0.350; M_1 0.726/0.211/0.606; M_2 0.923/0.103/0.340 (6,727 runs, 2,269 units).

Task identity alone thus lands inside the published 0.85–0.94 band; the whole trajectory adds +0.029 [+0.023, +0.034], improves log loss slightly (+0.0101), and leaves Brier unchanged (+0.0007, [−0.002, +0.003])—most visible gain is in ranking rather than probability refinement. Within task, M_0 is 0.5 by construction; the trajectory features alone reach 0.587 on this corpus (which pools all seventeen models, including the weak ones whose failures §5.10 shows are legible); a within-task value for M_2 is not identifiable under leave-one-out task scoring (the caution of §4.1).

Unlike the ICC bound, this measures what a predictor *gains*, on the same footing as the accuracies the literature reports.

5.4 Replication across scaffolds and domains

C2 (SWE-rebench / OpenHands; 67,074 trajectories over 1,823 repositories, failure rate 0.521) produces the cleanest null in the study: A_{within} is flat at 0.4935–0.5014 from $k=2$ to $k=20$ with every interval covering 0.50, the median unit's own AUROC exactly 0.500, and only 46–48% of units above chance—a null's signature, not a weak effect's—while the gap holds at +0.16 to +0.18 (pooled 0.660–0.672; difficulty oracle 0.974).

A difference between corpora. C1's within-task accuracy rises with depth (0.517 $\rightarrow$ 0.653) where C2's does not (0.494 $\rightarrow$ 0.497); re-running C1 with C2's exact eight features changes its curve by at most 0.035, so this tested featurisation does not explain the divergence. One explanation is consistent with §5.10: C2's single acting model is Qwen3-Coder-480B-A35B (per the corpus card), far stronger than C1's Llama-3 fine-tunes, and within-task legibility is negatively associated with capability in C3.

Hypothesis tests. Permutation tests of $H_0: A_{\text{within}} = 0.5$ (§2.5) reject in no condition after Holm: C2's null is tight (± 0.0057 ; a true 0.511 would flag) with observed 0.4934, and C3's one nominal hit ($k=5$, $p = 0.027$) misses the threshold. On C3's own pooled metric the label-only predictor beats the trajectory predictor by +0.134 [+0.120, +0.148]: **a predictor that never observes the agent wins on the field's metric** while exactly at chance on the one a monitor needs (tables in supplement).

5.5 Do hidden-state probes recover the signal?

The strongest objection to a behavioural null is that the signal is latent in the acting model's internal state [24, 28]. We test it on C4 by teacher-forced replay from the exact stored token ids, probing layers {10, 20, 30, 40} at prefixes $k \in \{1, 3, 5, 10\}$ with cross-fitted logistic probes (16 configurations per model). Across the 16 cells on C4-Q (174 mixed units) the mean within-task AUROC is 0.509 (median 0.513, range 0.458–0.551); one cell reaches 0.551 (Holm-adjusted $p = 0.048$; 95% CI [0.503, 0.588], at the resolution boundary of §2.5). On C4-L no cell survives (best $p = 0.40$, mean 0.513). The probes never beat a three-feature length baseline (+0.043, 95% CI [−0.014, +0.099]), while that same baseline attains the highest *pooled* AUROC of any predictor we test at every depth—and the activations demonstrably encode length (R^2 up to 0.98) and repository identity (AUROC up to 1.00), so what little the probes recover is plausibly length re-expressed. Probes over these representations reveal no strong early within-task signal, and the probe class is not the reason: a one-hidden-layer MLP moves the estimate by under 0.005 (0.519/0.526 at $k=1/10$) and mean-pooling states across cut points gives 0.484. Modest effects near the resolution boundary of §2.5 cannot be excluded. The full 16-cell grids with controls are in the supplement.

Can this pipeline detect an effect that is there? Running the identical pipeline on known answers: an outcome-carrying feature is recovered at within-task 1.000, 32 Gaussian-noise features give 0.503, and—the load-bearing check—*real activations with shuffled labels* give 0.488. The machinery reporting 0.524 on real labels returns 1.0 when signal exists and 0.5 when not; the estimator matches an independent implementation to 10^{-9} , and the permutation null centres at 0.502 ± 0.021 .

5.6 Trajectory length as the dominant signal

On pooled evaluation the length baseline does not merely compete—it wins. Three scalars (absolute token position, its log, assistant-turn count) beat the best 2048-dimensional activation probe at every depth on C4-L: pooled 0.710/0.712/0.707/0.706 against 0.639/0.619/0.652/0.677 at $k = 1, 3, 5, 10$ (142 units, full table in the supplement). Its *within*-task value is 0.502–0.503: length carries strong between-unit information while providing essentially no within-unit discrimination—this paper's point in miniature.

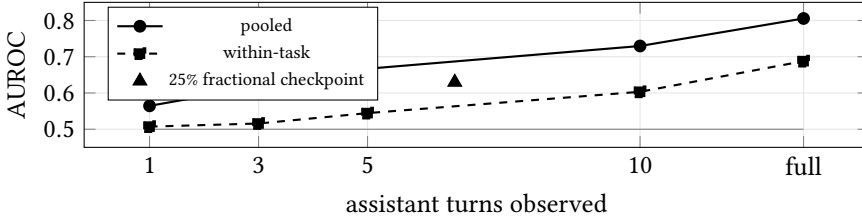


Fig. 2. The reimplemented published monitor on C1 (79,929 trajectories, 926 mixed units). Pooled AUROC overstates within-task discrimination by +0.06–0.13 at every depth; run-level signal is within resolution of chance at one turn and real only late (“full” plotted with an axis break at the right). The fractional checkpoint (triangle) sits above the within-task curve at its own observed depth—the leakage of Figure 3.

5.7 Bounding the null: equivalence and power

On the full C4-L sample (142 mixed units) the bootstrap SE of A_{within} is 0.0193; the 90% interval [0.491, 0.556] establishes equivalence to chance at $\delta = 0.10$ but misses $\delta = 0.05$ by 0.006, and the minimum detectable effect at 80% power is 0.054: we can exclude $A_{\text{within}} \geq 0.554$ and cannot exclude 0.53. The inflation gap itself is significant and tightens with sample: $A_{\text{pool}} - A_{\text{within}} = +0.107$, 95% CI [+0.046, +0.165]. The weighting of A_{within} does no hidden work: on C4-L unit-uniform macro-averaging gives 0.5243, task-demeaned 0.5173, min-repeat thresholds [0.5229, 0.5241]; and the pattern holds on the other corpora at $k=10$ —C1 pair/macro/deployment-weighted 0.587/0.581/0.576, C2 0.499/0.495/0.495—every variant on the same side of chance as the reported one.

5.8 Is the null a label-noise artifact?

Our labels are the benchmarks’ own oracles, and Wang et al. [33] report that 7.8% of “solved” SWE-bench patches fail the developer suite (6.2pp resolution inflation). Noise attenuates AUROC toward 0.5. Because a deployed pipeline is *trained* on the corrupted labels, attenuation compounds beyond the fixed-scorer formula (0.15 AUROC at $\alpha = 0.08$ where that formula predicts 0.03), so we calibrate it empirically: a true 0.55 falls to our observed 0.524 at $\alpha = 0.08$, almost exactly the documented mislabelling rate; a true 0.60 requires $\alpha = 0.49$; 0.65 never reaches it. Effects of 0.60 and above are excluded; effects near 0.55 are not, converging with the power analysis by an independent route. One property survives: attenuation is multiplicative in $A - \frac{1}{2}$, so noise can shrink a real effect but cannot manufacture our inflation gap (sweep tables in supplement).

A verification protocol for probe studies. Four gates, each of which caught a silent pipeline failure: (1) weight integrity (zero missing keys, teacher-forced accuracy floor 0.55—one code path silently randomised all 39 MoE layers; Apple-MPS loaded correctly yet computed wrong values); (2) replay from stored token ids; (3) an independent estimator agreeing to 10^{-9} ; (4) the positive-control battery above (supplement).

5.9 Re-evaluating a published predictor

Perhaps published methods reaching 0.94 have run-level signal our predictors miss. We reimplement Automata [5] from its description (artifacts unreleased; most of their corpora lack repeated runs, Table 1): activity abstraction, Last-Activity-Merge FSM with rare-transition filtering, gradient boosting over per-state behavioural plus cross-entropy anomaly features; 35–38 states, inside their 7–43; transition model fit on training folds only.

The reimplementation reproduces their reported result. Cho et al. [5] are explicit that their headline 0.941 is tau2-bench telecom; for SWE-agent, C1’s scaffold, they report 0.799. Our reconstruction

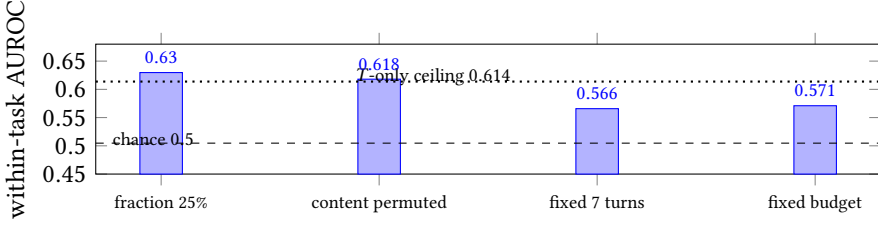


Fig. 3. Look-ahead leakage in fractional checkpoints (C1). The two leak-free protocols agree near 0.57; the fractional protocol scores 0.63 while observing slightly *less*, and keeps 0.618 even with content decoupled from outcomes within every unit—the surplus is the leaked final length, whose own ceiling is 0.614.

reaches **0.806** pooled on full C1 trajectories—within 0.007, same scaffold, FSM size inside their reported range—so the 0.941 a reader might expect us to match was never a SWE-bench number. (At the 25% checkpoint they report rank-AUROC 0.66 and we obtain pooled 0.728; different statistics, so no reproduction is claimed there.)

Because A_{pool} and A_{within} are computed from the *same* cross-fitted scores, their relationship is internally valid for our reconstruction; a different implementation could change either quantity and therefore the gap’s magnitude. Reproducing the published SWE-agent pooled figure closely reduces, but does not eliminate, that limitation. We would welcome the comparison on the authors’ own scores; the obstacle is data structure, since A_{within} needs repeated attempts, which most of their corpora do not record.

Early, essentially no run-level signal (Figure 2): at one turn $A_{\text{within}} = 0.507$ vs $A_{\text{pool}} = 0.565$ —within resolution of chance exactly where early abort would be worth most and the published metric says least.

Late, the signal is real: 0.603 by ten turns, 0.688 at full trajectory. Any claim that run-level prediction is impossible would be false; what the pooled figure obscures is *when* it becomes possible.

The gap persists at every depth, between +0.057 and +0.126. Pooled AUROC overstates run-level discrimination throughout, and by an amount that grows as the predictor gets better.

The fractional checkpoint leaks, and we can measure how much. The triangle in Figure 2 is the paper’s own online protocol, evaluated at 25% of the trajectory. Under it the monitor observes a mean of 6.7 assistant turns (median 4, IQR 2–8)—yet scores higher than fixed-turn protocols observing as much or more. The higher score at a smaller average prefix is a warning sign, not itself proof; the matched controls below identify the source as *look-ahead leakage*: a checkpoint defined as a fraction of the trajectory depends on its eventual total length T , unavailable online and itself outcome-associated.

We measure the leak three ways (Figure 3). *Ceiling:* T alone achieves within-task 0.6138 [0.600, 0.627]. *Matched control:* a fixed-turn protocol observing marginally *more* (6.8 vs 6.6 turns) scores 0.5658 vs 0.6297—the +0.064 difference is the leak. *Source:* length-only reaches 0.6268 but content-only also 0.6255 (content over a T -proportional window encodes T); the separation is a within-unit permutation, which decouples content from outcome while each run keeps its length and still leaves 0.6180—destroying every content–outcome relationship costs 0.011.

None of this is a criticism of the method. It is a property of the evaluation protocol and applies to any system evaluated at fractional checkpoints—including both prior negative results [16, 17], which evaluate at 50% and at 10–50% of trajectory respectively, so by this measurement their early figures are if anything optimistic. The protocol matters (the supplement states this as a proposition: any cut index that is a function of final length makes T learner-recoverable, lower-bounding attainable

within-task AUROC by T 's own 0.614; fixed-index protocols admit no such bound), and the remedy is cheap: cut at a fixed number of turns or a fixed token budget, both computable without knowing the future. On C1 these give $A_{\text{within}} = 0.5658$ and 0.5710 respectively—close to each other and far below the fractional protocol's 0.6297, which is what a leak-free protocol should look like.

5.10 Depth, capability, and what the probe null means

Signal accumulates with depth, and it is not selection. Repeating the sweep with the automaton featuriser on a *fixed cohort*—the 63,657 C1 traces with ≥ 10 turns, identical at every depth—gives 0.509/0.511/0.532/0.586 at $k = 1, 3, 5, 10$ (885 units; varying-sample values agree within 0.007; Fig. 2's curve differs slightly by including shorter traces whole). Early discrimination is within resolution of chance on a constant trace set, clearly above it by ten turns; Table 3's 0.599 at $k=10$ uses the behavioural+TF-IDF predictor, a different instrument. Survivor selection can only inflate a curve that *rises*: among our corpora that is C1 (C2 and the C4 probe are flat with depth, C3 has fixed three runs).

Legibility falls as the acting model strengthens. Across C3's 17 models (one predictor, cross-fitted by task, evaluated per model): $\rho = -0.662$ ($p = 0.0038$); the weakest model's failures are identified at 0.832, the two strongest at 0.444 and 0.490. Within lineage, 15 of 19 same-family pairs show the stronger member less legible (dose-response $\rho = -0.749$, $p = 0.0002$), though 15 of those pairs come from one family. The association survives a battery of checks: permutation, matched pair counts, capability measured on disjoint tasks and on disjoint domains, leave-one-family-out, and a precision-weighted slope of -0.674 ± 0.113 , and a model-resample 95% CI of $[-0.90, -0.23]$ that accounts for tasks shared across models (supplement). The direction also replicates on C1's second scaffold, where per-model within-task AUROC at ten turns orders Llama-8B 0.636 > 70B 0.581 > 405B 0.569 (the outer two on 74 and 44 units). It remains an association on one primary scaffold, not a causal claim: run-level legibility declines with capability in these data, suggesting runtime prediction *may* grow harder as models strengthen.

The obvious explanation does not hold. Behavioural dispersion (outcome-blind within-unit Jensen–Shannon divergence [1]) does fall with capability ($\rho = -0.591$), echoing Gould's shrinking-variance account [12]—but partialling it out changes nothing ($-0.662 \rightarrow -0.668$). Instead the signal vanishes from every channel at once: in the weakest tercile, surprise, output size and action identity each separately identify failure (0.61–0.64); in the stronger terciles no family clears chance. This pattern is consistent with weak agents exposing failure through coarse procedural cues while stronger agents' failures are less visible to the behavioural summaries we measure; the underlying failure mechanism remains open (full tables in the supplement).

This reconciles the corpora and the probe. C2's single, much stronger model (Qwen3-Coder-480B-A35B) stays at chance at every depth (0.500–0.504; 1,741 mixed units, 38,576 pairs, a well-powered null with no selection, every trace exceeding ten turns) where C1's weaker Llamas climb to 0.586. And the C4 probe null is the *expected* observation: featurising C4's recorded trajectories exactly as C1–C3 (Table 5), behaviour is also at chance at the probed depths (0.501/0.496/0.514; 192 mixed units, no token cap) and reaches 0.656 only at full trajectory. Nothing is legible to either channel that early. A monitoring policy should not assume predictability transfers from another deployment; it must be established for one's own model and scaffold.

6 WHAT TO DO INSTEAD

A critique that stops at “the metric is wrong” leaves practitioners nothing; this section gives a diagnostic to report and a policy that performs strongly in replay.

Table 5. Within-task discrimination at matched prefix depths. First three rows use the automaton behavioural featuriser; the last is the best of four hidden-state probe layers per depth (generous to the probe).

	$k=1$	$k=5$	$k=10$	full
C1 (Llama 8B-405B)	0.509	0.532	0.586	0.688
C2 (single stronger model)	0.504	0.501	0.503	—
C4-L, behaviour	0.501	0.496	0.514	0.656
C4-L, activations	0.524	0.527	0.546	n/a

6.1 Report the gap

Report $(A_{\text{pool}}, A_{\text{within}}, w)$ and the gap $\Delta = A_{\text{pool}} - A_{\text{within}}$: $\Delta \approx 0$ means the predictor reads runs; $\Delta \approx A_{\text{pool}} - 0.5$ means it reads unit identity and nothing else. Here Δ ranges +0.09 to +0.47; it costs nothing beyond runs a repeated-attempt benchmark already has.

6.2 A Bernoulli-arm reduction

Within a unit: no run-level signal above resolution (§5.5); cost independent of outcome given task ($p = 0.18, 0.63$; paired difference $< 0.7\%$ of mean cost); ten temporal diagnostics (lags 1–3 against a permutation null carrying the $-1/(n-1)$ bias, runs tests, half-splits) show one marginal signal ($p = 0.045$ uncorrected). None of this *proves* exchangeability; we treat exchangeable Bernoulli(θ_u) outcomes as an approximation the diagnostics do not contradict, and evaluate by replaying real runs, not the model. Under it, allocation is a budgeted multi-armed bandit: each task an arm, each pull costing random c_u , a token budget B limiting pulls, an arm retiring on success. Retirement is endogenous—outside both the budgeted bandit [35] and mortal-bandit [4] analyses—so no regret guarantee is claimed. In the fluid limit, expected cost to solve u is $\bar{c}_u/\theta_u$, so greedy $\theta_u/\bar{c}_u$ maximises solves under budget—a ratio-index policy (cf. Smith’s rule [29]) the oracle implements and Thompson sampling approximates (supplement).

6.3 Budgeted Thompson sampling with a transferred prior

Budgeted Thompson Sampling [35], Beta-Bernoulli: sample $\tilde{\theta}_u$ per arm, pull $\arg \max_u \tilde{\theta}_u/\bar{c}_u$, update. The prior transfers another model’s outcomes on the same task ($\alpha_u = 1 + w s_u^{\text{other}}, \beta_u = 1 + w(n-s_u^{\text{other}})$, $w=0.5$). Replay evaluation: each pull draws a real recorded run, charges its real cost, returns its real outcome, retires solved tasks; 200 budget realisations. The contribution is reduction, prior and evaluation, not the algorithm.

6.4 Does it work?

The effect is large exactly where theory predicts: under budget pressure. On the **C4-Q target** (prior transferred from C4-L; full table in the supplement) the policy solves 162.5 tasks at 2M tokens against round-robin’s 53.1— $3.1\times$ for the same spend, 94% of an oracle knowing every true success rate—with the transferred prior contributing +14–20% over an uninformative one. Table 6 reports the **C4-L target** under the same replay semantics (109.5 vs 30.5 at 2M, i.e. $3.6\times$): the two headline multipliers are the two transfer directions, not a discrepancy, and “round-robin”/“uniform” name the same policy on the two splits. The margin narrows with budget and vanishes by 40M, where allocation stops mattering—a sanity check, since gains in the unconstrained regime would be suspect.

Table 6. Distinct tasks solved at a fixed token budget on C4-L, 150 replay trials. ALLOCATE is budgeted Thompson sampling over tasks; HYBRID adds aborting to it.

Budget	Round-robin	Turn-cap 10	Abort@5	ALLOCATE	HYBRID
2M	30.5	2.6	34.9	109.5	104.2
5M	75.8	5.8	87.1	194.4	181.0
10M	151.3	9.2	165.8	276.8	248.9
20M	291.2	12.4	236.8	332.8	292.8

6.5 Ablations

A *static* transferred-prior ranking matches the bandit at 2M but collapses beyond (208.0 flat)—a partly wrong prior never corrects without the online component. Give-up and cost-sampling variants change counts by under one task; reversing transfer direction preserves the result (+195%/+117% at 2M/5M). Tables in supplement.

6.6 Where the prior comes from

Difficulty transfers across models ($r = 0.863$; a transferred prior keeps 92% of oracle ranking power), so the policy survives upgrades. For genuinely novel tasks, pre-execution prediction is real but not yet actionable: Ge et al. [11]’s released features predict our observed difficulty at $\rho \approx 0.53$ held out (rollout-derived references: 0.83–0.85), yet seeding the prior from them *loses* to an uninformative prior at every weight (−0.3% to −11.1%) while the rollout-derived prior gains +14.4%. Between $\rho = 0.53$ and 0.85 lies a usefulness threshold: pre-execution estimation serves triage (Q1), not allocation (supplement).

6.7 Abort or reallocate? A direct comparison

Reviewers will reach for the obvious baselines—cap the turns, or abort on a monitor’s verdict—so we implement them under one budget on C4-L. (Turn control is an active SE efficiency line—Gao and Peng [10] cut coding-agent cost with fixed/dynamic turn budgets and no predictor; our turn-cap row is the replay-feasible instance, and allocation operates orthogonally at the task level.) Replay as in §6.3; aborting at turn k charges only tokens consumed through k (generous accounting: any forgone late success is free). The monitor is the cross-fitted automaton predictor at $k=5$, threshold at the median prefix score—a monitor of exactly the measured quality.

Three results in Table 6. Naive aborting helps slightly at scarce budgets (+14% at 2M, a budget effect: cheaper pulls touch more tasks) and hurts otherwise (−19% at 20M). A hard turn cap is the worst policy on the board (2.6–12.4 solved): only 0.8% of successful C4-L runs finish within ten commands, so the cap destroys almost every success it was protecting.

The monitor’s operating curve—what a practitioner reads, since a ranking metric fixes no operating point—makes this concrete. Sweeping the $k=5$ abort threshold on C4-L, to retain 99/97/95% of successful runs the monitor catches only 1.4/5.4/9.1% of failures and saves 1.2/4.3/7.3% of compute: keeping 95% of successes buys back 7% of spend—a flat curve where aborting would pay.

A fourth, from simulated abort-and-restart on C1 (supplement): the pooled-selected predictor harms end-to-end success at all seven operating points; the within-task-selected one is non-negative at all twelve (± 0.3 pp, both near break-even [31]). Metric choice changes which policy appears deployable.

How the score is used matters more than whether. A second replay experiment (120 trials; own baselines: round-robin 30.9, ALLOCATE 322.3 at 2M/20M—the latter slightly below Table 6’s because this run fixes the prior weight at 0.5) separates the score’s unit-level from its run-level value. *Routing*

on the $k=5$ score alone—as a noisy difficulty estimate, no outcome history for the routed task (the monitor is cross-fitted, trained only on *other* tasks’ outcomes), no aborting—solves 79.0 at 2M vs 30.9: in the novel-task regime the monitor is a useful *difficulty estimator* ($2.6\times$). But that value is unit-level and subsumed once real outcomes exist: sweeping the evidence weight, monitor input layered on the transferred prior never beats this experiment’s ALLOCATE (320.2 vs 322.3 at 20M at the lightest weight) and hurts monotonically as trusted more (to 142 at weight 1), while a *fair* abort (break-even of §2.4; in-sample calibration favouring it) recovers nearly all the naive loss yet still does not beat allocation (317.0 vs 322.3)—“aborting adds nothing” is thus conservative. In short: route with it when nothing better exists; calibrate any abort to break-even; never let it override observed outcomes.

6.8 Limitations of the proposal

The method needs prior attempts (§6.6 bounds the novel case); gains vanish by 40M tokens; assumptions are tested approximations (§6.2); held-out weight selection moves throughput 0.5%; no regret guarantee (endogenous retirement [4, 35]); replay cannot capture live interactions.

7 IMPLICATIONS

For researchers: report within-task accuracy and the same-unit weight beside the pooled figure, and include two cheap confound-exposing baselines—task difficulty and trajectory length. *For monitor designers:* use prefixes computable online (never fractions of an unknown future), establish conditional predictability for your own model and scaffold, and ship positive controls (silent probe failures here were caught only by the gates of §5.8). Monitoring is not worthless—signal exists late and for weaker models—but the pooled metric cannot say whether it exists for yours.

8 THREATS TO VALIDITY

Construct. Labels are the benchmarks’ oracles (7.8% of “solved” patches fail the developer suite [33]; 10.7% of passes flawed [25])— outcome-label noise is a known confounder of SE prediction models [30], bounded here in §5.8; some monitors target process quality.

Internal. Runs may not be exchangeable (ten diagnostics; one marginal $p = 0.045$). The probe’s 16,000-token cap excludes the 129/500 longest C4-L units; the estimate is stable as longer units enter ($0.498 \rightarrow 0.513$) and the cap-free behavioural analysis (192/203 mixed units) agrees at probed depths. Our reimplementations could differ from the original despite matching its number.

External. Corpora are coding-heavy; C3’s eight non-coding domains still show the oracle at pooled 0.909, so between-unit domination is not code-specific. The probe corpus is one benchmark family and two models, the capability association one primary scaffold, and allocation replay one benchmark family.

Statistical. A_{within} is identifiable only on mixed-outcome units (21.9–40.6%), a principal stratum [9]; sensitivity checks show no inflation as the stratum widens. In multi-model corpora (C1, C3) units share tasks, so we report the capability trend under a model-resample bootstrap; the single-model probe splits (C4) and C2 have no shared-task dependence. Probe resolution ≈ 0.55 (§2.5); grids Holm-corrected.

9 CONCLUSION

Agent failure predictors report AUROC of 0.85–0.94, read as evidence a running agent can be monitored. That number answers a different question: pooled AUROC has same-unit weight $\Theta(1/n)$ (0.003–0.07% here) with a difficulty-only ceiling predictable from the outcome table, and a leave-one-run-out oracle reaches 0.894 while unable to tell two attempts apart. On the estimand a monitor needs, early discrimination is weak across four corpora; a reproduced predictor reaches 0.688 only at full trajectory (not selection), and its fractional protocol leaks length worth +0.064. Within C3, legibility declines as models strengthen ($\rho = -0.66$). Pooled evaluation cannot say whether a monitor supports intervention: in replay, naive aborting costs up to 19% of solved tasks while routing the same signal multiplies solves—so the estimand a benchmark reports should be the one its deployment decision needs.

Data availability. Scripts, metric tests, supplementary tables, a make reproduce-main pipeline, and corpus names/licences are in the replication package at <https://github.com/seanonymoussubmission/agent-monitor-estimand> (the account is identity-free for double-blind review).

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